

VFQEC E1 - aca95affb4e946598029f8c307c01b80

BlueQubit cpu coherent simulation + local noise/syndrome/recovery

Method	Final error	95% interval	Fitted / round
Unencoded qubit	0.25725	[0.2439, 0.271]	0.0873256
Standard QEC	0.0055	[0.003635, 0.008314]	9.4078e-05
QEC + twirling	0.004	[0.002464, 0.006488]	8.59271e-05
QEC + calibrated field	0.0005	[0.0001371, 0.001821]	1.26067e-05
QEC + adaptive field	0.0005	[0.0001371, 0.001821]	1.02507e-05
Oracle (single-qubit field)	0.0005	[0.0001371, 0.001821]	9.9054e-06

Phenomenological data noise; ideal gates, preparation and terminal readout.

Spatial decoding each round; $q > 0$ is not fault-tolerant spacetime decoding.

Calibration uses fresh encoded probes; drift time freezes during calibration.

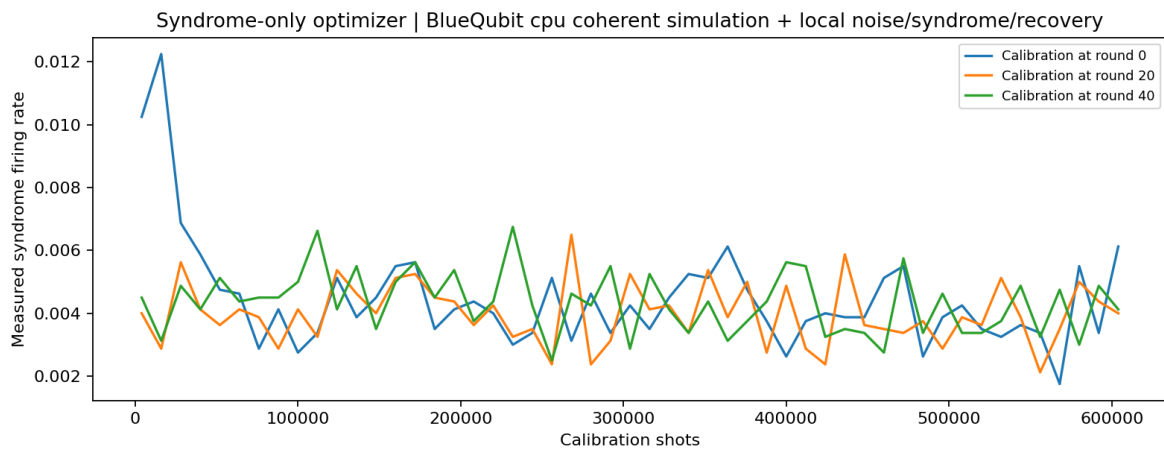
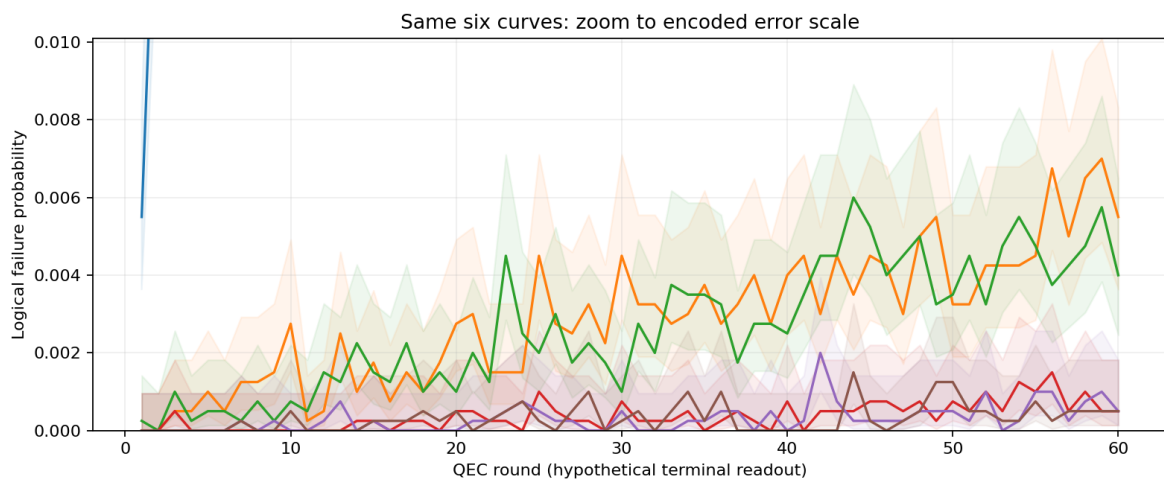
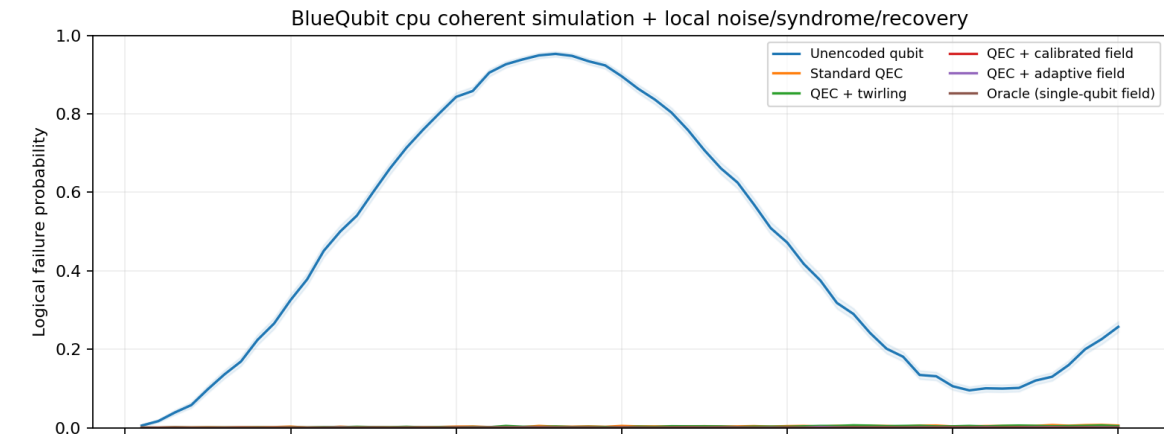
Oracle cancels single-qubit fields only; ZZ and stochastic noise remain.

Endpoint curves share simulated trajectories and have correlated time points.

Zero observed failures have nonzero confidence upper bounds.

Rates are descriptive; coherent/drift noise need not be exponential. No fit covariance is interpreted as independent-round uncertainty.

unencoded: exponential model poorly describes coherent/drift memory



Evaluation-only ground truth; dotted = calibrated | BlueQubit cpu coherent simulation + local noise/syndrome/recovery

